METRIC

MIL-DTL-15090F
16 January 2026
SUPERSEDING
MIL-DTL-15090E(SH)
w/INT. AMENDMENT 1
10 September 2013
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DETAIL SPECIFICATION

ENAMEL, EQUIPMENT, LIGHT GRAY (NAVY FORMULA NO. 111)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers enamel that has been formulated to comply with air pollution regulations that allow a maximum volatile organic compound (VOC) content of 250 grams of solvent minus water per liter of paint (2.08 lbs/gal) as delivered. The paint is an air-drying, solvent-based, light-gray equipment enamel (Navy Formula no. 111) for shipboard use as a topcoat on equipment, machinery, furniture, and electrical equipment, such as switchboards, including applications covered by reactor plant painting schedules. The enamel color is also cited as a bulkhead gray, machinery gray, or bulkhead machinery gray in procurement requirements. This product is supplied in a single can and is used as supplied without thinning.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-P-645 - Paint, Alkyd Type, Formula Number 84

FEDERAL STANDARDS

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680 - Degreasing Solvent

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-PRF-22262	-	Abrasive Blasting Media, Ship Blast Cleaning
MIL-DTL-24441	-	Paint, Epoxy-Polyamide, General Specification for
MIL-PRF-24596	-	Coating Compounds, Nonflaming, Fire-Resistant
MIL-DTL-24607	-	Enamel, Interior, Nonflaming (Dry), Chlorinated Alkyd Resin
MIL-PRF-24635	-	Coating Systems, Weather-Resistant, Exterior Use
MIL-PRF-24763	-	Enamel, Emulsion Type, for Shipboard Use

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910.1200	-	Hazard Communication
29 CFR 1910.1200, Subpart Z	-	Toxic and Hazardous Substances
29 CFR 1915.1001, Appendix K, Method ID-191	-	Polarized Light Microscopy of Asbestos – Non Mandatory
40 CFR 60, Appendix A-7, Method 24	-	Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings
40 CFR 63, Subpart II	-	National Emission Standards for Shipbuilding and Ship Repair (Surface Coating)
40 CFR 82	-	Protection of Stratospheric Ozone
40 CFR 261.24(a), Method 1311	-	Toxicity Characteristic Leaching Procedure (TCLP)
40 CFR 355, Appendices A and B	-	The List of Extremely Hazardous Substances and Their Threshold Planning Quantities
40 CFR 372.65, Subpart D	-	Specific Toxic Chemical Listings

(Copies of these documents are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010	-	Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

(Copies of the chapter titled “Material Control Program” are available by email request to CommandStandards@navy.mil.)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 600/4-79-020	-	Methods for Chemical Analysis of Water and Wastes
EPA Method 311	-	Analysis of Hazardous Air Pollutant Compounds in Paints and Coatings by Direct Injection into a Gas Chromatograph
EPA SW-846	-	Test Methods for Evaluating Solid Waste, Physical/Chemical Methods

(Copies of these documents are available online at www.epa.gov.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D185	-	Standard Test Methods for Coarse Particles in Pigments
ASTM D476	-	Standard Classification for Dry Pigmentary Titanium Dioxide Products
ASTM D522/D522M	-	Standard Test Methods for Mandrel Bend Test of Attached Organic Coatings
ASTM D523	-	Standard Test Method for Specular Gloss
ASTM D562	-	Standard Test Method for Consistency of Paints Measuring Krebs Unit (KU) Viscosity Using a Stormer-Type Viscometer
ASTM D823	-	Standard Practices for Producing Films of Uniform Thickness of Paint, Coatings, and Related Products on Test Panels
ASTM D869	-	Standard Test Method for Evaluating Degree of Settling of Paint
ASTM D1210	-	Standard Test Method for Fineness of Dispersion of Pigment-Vehicle Systems by Hegman-Type Gage
ASTM D1364	-	Standard Test Method for Water in Volatile Solvents (Karl Fischer Reagent Titration Method)
ASTM D1475	-	Standard Test Method for Density of Liquid Coatings, Inks, and Related Products
ASTM D1849	-	Standard Test Method for Package Stability of Paint
ASTM D2244	-	Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates
ASTM D2369	-	Standard Test Method for Volatile Content of Coatings
ASTM D2698	-	Standard Test Method for Determination of the Pigment Content of Solvent-Reducible Paints by High-Speed Centrifuging
ASTM D2805	-	Standard Test Method for Hiding Power of Paints by Reflectometry
ASTM D3278	-	Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus
ASTM D3359	-	Standard Test Methods for Rating Adhesion by Tape Test
ASTM D3363	-	Standard Test Method for Film Hardness by Pencil Test
ASTM D4400	-	Standard Test Method for Sag Resistance of Paints Using a Multinotch Applicator
ASTM D5150	-	Standard Test Method for Hiding Power of Architectural Paints Applied by Roller
ASTM D5895	-	Standard Test Methods for Evaluating Drying or Curing During Film Formation of Organic Coatings Using Mechanical Recorders

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ASTM E1252 - Standard Practice for General Techniques for Obtaining Infrared Spectra for Qualitative Analysis

ASTM E1347 - Standard Test Method for Color and Color-Difference Measurement by Tristimulus Colorimetry

ASTM F718 - Standard for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-STD-595 - Colors Used in Government Procurement

SAE-AMS-STD-595/26307 - Bulkhead Machinery Gray, Semigloss

(Copies of these documents are available online at www.sae.org.)

THE ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC-SP 10/NACE No. 2 - Near-White Metal Blast Cleaning

(Copies of this document are available online at www.ampp.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Formula. The enamel shall consist of an alkyd resin, pigments, extender pigments, and solvent, as described in sections 3.2.1 and 3.2.2, that when combined, shall result in a paint conforming to the requirements of the applicable paragraphs of this specification (see 6.5).

3.2.1 Navy Formula no. 111. The formula shown in table I is designated Navy Formula no. 111. When Navy Formula no. 111 is specified, or when this specification is referenced without reference to the formula number, the enamel shall conform to the requirements of this specification. Alternate ingredients may be used as long as the alternate ingredients are specifically approved by Technical Authority (see 6.13) and the enamel is shown to meet all the requirements of this specification by first article testing (see 6.4) and conformance testing. Conformance testing may be conducted by the Government at any time to verify that the product being acquired meets the requirements of this specification (see 4.3).

TABLE I. Navy Formula no. 111.

Ingredients	Kilograms (kg)	Pounds (lb)	Percentage by weight (%WT)
Alkyd resin solution 1/	291.44	642.52	56.41
Titanium dioxide 2/	56.23	123.96	10.88
Magnesium silicate 3/	62.97	138.83	12.2
Parachlorobenzotrifluoride 4/	45.22	99.7	8.75
Paint thinner 5/	36.35	80.15	7.04
Suspending agent 6/	9.0	19.83	1.74
Dispersant 7/	1.8	3.96	0.35
Calcium drier 8/	4.9	10.81	0.95
Zirconium drier 9/	2.45	5.4	0.47
Antiskinning agent 10/	0.9	1.98	0.17

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TABLE I. Navy Formula no. 111 – Continued.

Ingredients	Kilograms (kg)	Pounds (lb)	Percentage by weight (%WT)
Cobalt drier 11/	0.73	1.62	0.14
Yellow iron oxide 12/	2.70	5.95	0.52
Antifloat 13/	1.35	2.98	0.26
Carbon black 14/	0.63	1.4	0.12
Totals 15/	516.67	1,139.1	100.0

NOTES:

1/ The alkyd resin complies with the table II requirements.

2/ ASTM D476, types I or II.

3/ Talcron MP-1250, Specialty Minerals, Inc., Bethlehem, PA, or equal.

4/ Parachlorobenzotrifluoride (PCTBF), CAS # 98-56-6.

5/ Mineral Spirits (see MIL-PRF-680).

6/ Bentone 38 (10%) Gel, NCP Coatings, Niles, MI, or equal.

7/ Spurso, Borchers (Milliken); Westlake, OH, or equal.

8/ Calcium drier (6%), Calcium neodecanoate OMG Mooney was used in the formulation or functionally equivalent driers from Borchers (Milliken), Westlake, OH or equal may be used in the formulation

9/ Zirconium neodecanoate (18%) from OMG Mooney was used in the formulation or a functionally equivalent drier from Borchers (Milliken), Westlake, OH or equal may be used to achieve 0.30% Zr, by weight in the formulation.

10/ Methyl ethyl ketoxime (MEKO), CAS # 96-29-7, Exkin #2, was used for this formulation, or functionally equivalent CAS # 96-29-7 MEKO alternatives are available from other manufacturers and may be used in the formulation.

11/ Cobalt neodecanoate (12%) from OMG Mooney was used in the formulation, functionally equivalent cobalt drier (12%) from OH Borchers (Milliken), Westlake, OH or equal may be used to achieve 0.060% Co, by weight in the formulation.

12/ Y25-LO from Elementis Pigments was used in the formulation, or functionally equivalent alternatives of CAS # 51274-00-1 YO-3587 from Huntsman Advanced Materials Americas, The Woodlands, TX or equal may be used in the formulation.

13/ Troysol AFL, antifoam and leveling additive, Troy Corporation (Arxada), Newark, NJ.

14/ Raven 14, Birla Carbon U.S.A. Inc, Marietta, GA, or equal.

15/ This formula makes approximately 379 liters (100 gallons) of paint.

3.2.2 Composition. The paint shall consist of the ingredients listed in table I footnotes in the proportions shown in table I and table II except that the amount of carbon black may be varied as necessary to meet color requirements. In addition, to assist in meeting drying times, brushing, condition in container, and storage stability requirements, driers, antisettling, antisagging, and antiskinning agents may be varied as necessary to meet the other requirements of this specification.

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TABLE II. Alkyd resin solution quantitative requirements.

Characteristics [1/]	Requirements	
	Minimum	Maximum
Nonvolatile resin, %WT of solution	49	51
Phthalic anhydride, %WT of non-volatile vehicle	30	-----
Drying oil acids, %WT of non-volatile vehicle	45	55
Acid number	5	12
Dihydric alcohol, %WT of non-volatile resin	-----	7
Viscosity (G-H) at 25 °C	U	Z
Color, gardener	-----	10
Weight per gallon, kg/liter (pounds/gallon)	1.16 (9.7)	1.28 (10.7)
Soya oil	Positive (by certification) [2/]	
Rosin	Negative (by certification) [2/]	
Phenolic resin	Negative (by certification) [2/]	

NOTES:

1/ These characteristics describe a 50% (medium oil), soya-based, alkyd resin solution in a PCTBF carrier solvent that is free of rosin or phenolic modification.

2/ Certification requirements for the formulator shall be to receive a confirmation from the manufacturer of the composition of the product. This may be either outlined in the product's technical data sheet or in a certificate of analysis provided upon receipt of the product.

3.2.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Toxicity. When evaluated in accordance with 4.7, the enamel shall pose no serious or high risk to the health of personnel or the environment for its intended purpose (see 4.7 and 6.8). The materials content used in the enamel shall have no known reportable carcinogenic or potentially carcinogenic materials and shall have no extremely hazardous substances (EHS) or toxic chemicals identified in 40 CFR 355 and 372.65, respectively. The enamel shall not contain more than 0.005 percent cobalt compounds by weight in its formulation, unless a cobalt drier is used (3.5).

3.4 Manufacture. The component raw materials shall be mixed, ground, and blended as required to produce a product that is uniform, free from grit, lumps, skins, and floating pigment.

3.5 Quantitative requirements. The enamel shall conform to the quantitative requirements described in table III and 3.5.1, 3.5.2, and 3.6.

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TABLE III. Navy Formula no. 111 quantitative requirements.

Characteristics	Requirements	
	Minimum	Maximum
Pigment, %WT of enamel	22.0	25.0
Volatiles, %WT of enamel	46.0	49.0
Nonvolatile vehicle, %WT of enamel (calculated by difference)	28.0	31.0
Water, %WT of enamel	------	0.5
Coarse particles and skins (as residue on N. 325 sieve) %WT of enamel	------	0.2
Viscosity, Krebs units	75	105
Weight per gallon, kg/liter (pounds/gallon)	1.31 (10.9)	1.42 (11.9)
60° Specular gloss	35	50
Contrast ratio	0.97	------
Color, ΔE	------	0.5
Time to dry to touch (hours)	------	2.0
Time to dry through (hours)	------	8.0
Pencil hardness	HB 1/	------
Fineness of grind, Hegman units	5	------
Flashpoint, C (F)	38° (100°)	------
Sag resistance, mm (mils)	0.11 (5)	------
VOC, g/L (pounds/gallon)	------	250 (2.08)
Asbestos content, %WT of dry film	------	0.5
Crystalline silica content, %WT of dry film	------	0.5

NOTE:
1/ Hard black

3.5.1 Hazardous metal content. When tested as specified in 4.6.2, the content of each soluble metal and total content of each metal of the enamel shall be less than the values as specified in table IV and table V. If the test for total metal results in a value less than the soluble metal limit, the soluble metal test need not be conducted. The total and soluble metal values shall be reported in table IV and table V results.

TABLE IV. Hazardous soluble metals content.

Soluble metal and its compounds	Maximum, mg/L
Antimony and its compounds	15.0
Arsenic and its compounds	5.0
Barium and its compounds (excluding barite)	100.0
Beryllium and its compounds	0.75
Cadmium and its compounds	1.0
Chromium (VI) compounds	1.0

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TABLE IV. Hazardous soluble metals content – Continued.

Soluble metal and its compounds	Maximum, mg/L
Chromium and its chromium (III) compounds	560.0
Cobalt and its compounds	50.0
Copper and its compounds	25.0
Fluoride salts	180.0
Lead and its compounds	5.0
Mercury and its compounds	0.2
Molybdenum and its compounds	350.0
Nickel and its compounds	20.0
Selenium and its compounds	1.0
Silver and its compounds	5.0
Tantalum and its compounds	100.0
Thallium and its compounds	7.0
Tungsten and its compounds	100.0
Vanadium and its compounds	24.0
Zinc and its compounds	250.0

TABLE V. Total hazardous metals content.

Metal	Maximum %WT
Antimony and its compounds	0.015
Arsenic and its compounds	0.005
Barium and its compounds (excluding barite)	0.10
Beryllium and its compounds	0.0002
Cadmium and its compounds	0.0005
Chromium (VI) compounds	0.001
Chromium and its chromium (III) compounds	0.56
Cobalt and its compounds	0.005 1/
Copper and its compounds	0.01
Fluoride salts	0.18
Lead and its compounds	0.005
Mercury and its compounds	0.0002
Molybdenum and its compounds	0.35
Nickel and its compounds	0.02
Selenium and its compounds	0.002
Silver and its compounds	0.001

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TABLE V. Total hazardous metals content – Continued.

Metal	Maximum %WT
Tantalum and its compounds	0.100
Thallium and its compounds	0.007
Tungsten and its compounds	0.100
Vanadium and its compounds	0.01
Zinc and its compounds	0.25

NOTE:
1/ Total cobalt content may exceed 0.005 %WT (up to 0.2 %WT) only if a cobalt drier is used to effect proper drying. Regardless of whether a cobalt drier is used, soluble cobalt content shall not exceed table IV requirements.

3.5.2 Hazardous air pollutants (HAP). When evaluated as specified in 4.6.3, the total HAP content of the enamel, as defined by the 40 CFR 63 Subpart II, (EPA National Emission Standards for Hazardous Air Pollutants) requirements for shipbuilding and ship repair, the VOC content shall not exceed 250 g/L (2.08 lb/gal). The finished enamel shall meet all the requirements of this specification and the solvent selection shall comply with table I.

3.6 Qualitative requirements. The enamel shall conform to the qualitative requirements as specified in 3.6.1 through 3.6.10.

3.6.1 Odor. When tested as specified in 4.6, the odor shall be characteristic of the volatiles cited in table I.

3.6.2 Color. When tested as specified in 4.6.6, the color shall match color number 26307 in accordance with SAE-AMS-STD-595. The measured color deviation, ΔE, from color number 26307 of SAE-AMS-STD-595 in Commission Internationale de l'Eclairage (CIE; International Commission on Illumination) LAB units shall be not greater than 0.5.

3.6.3 Flexibility. When tested as specified in 4.6.7, a film of the enamel shall show no evidence of cracking or flaking.

3.6.4 VOC regulations. This paint shall be used as supplied without thinning.

3.6.5 Condition in container. When tested as specified in 4.6.8, a freshly opened, full container of enamel shall be free from lumps, abnormal thickening, or livering and shall show no more pigment settling or caking than can be readily reincorporated to a smooth, uniform state.

3.6.5.1 Storage stability.

3.6.5.1.1 Full container. When tested as specified in 4.6.9.1, the product shall be capable of being readily broken up with a paddle to a smooth uniform consistency. It shall be free from livering, curdling, gelling, pigment float, hard caking, gummy sediment, or other objectionable properties for at least 1 year after date of manufacture. The enamel shall conform in all respects to this specification.

3.6.5.1.2 Partially full container. When tested as specified in 4.6.9.2 and ASTM D1849, a three-quarter filled, closed 225-milliliter (8-ounce) glass jar of enamel shall show no skinning at the end of 48 hours, and after aging as specified, the enamel shall show no skinning, surface float (separated pigments or other ingredients on the surface of the liquid enamel), livering, curdling, hard caking, or gummy sediment. The enamel shall mix readily to a smooth, uniform state within 5 minutes when mixed with a paddle or mechanical mixer and shall have a consistency not greater than 5 units from the original Kreb unit consistency before testing (see table III). Enamel shall meet all other requirements of this specification.

3.6.6 Application properties.

3.6.6.1 Rolling properties. When tested as specified in 4.6.15.1 the enamel shall roll satisfactorily and shall dry to a uniform film, free from seed, runs, sags, or streaks. The dried film shall show an even, smooth finish.

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3.6.6.2 Brushing properties. When tested as specified in 4.6.15.2, the enamel shall be capable of being brushed out and laid off without excess drag on the brush. When dry, the brushed surface shall be free of sags and runs and shall show a minimum of brush marks.

3.6.6.3 Spraying properties. When tested as specified in 4.6.15.3, the enamel shall spray satisfactorily in all respects and shall show no running, sagging, streaking, or orange peel. The film shall air dry and show no seeding, dusting, floating, fogging, mottling, hazing, or other film defects.

3.6.7 Cross-cut adhesion test. When tested as specified in 4.6, 4.6.11.1, and 4.6.11.2, the dry film of enamel shall achieve a classification of 4A or 4B.

3.6.8 Water resistance. When tested as specified in 4.6.10, a film of enamel shall show no wrinkling or blistering immediately after removal of the panel from water. The enamel shall be only slightly affected with regard to adhesion when observed 2 hours after removal from the water. After 24 hours air drying the portion of the panel that was immersed shall be almost indistinguishable, with regard to adhesion (see 3.6.7) and general appearance (see 3.6.10), from the portion of the panel that was not immersed.

3.6.9 Compatibility.

3.6.9.1 Compatibility with topcoats. When tested as specified in 4.6.11.1, the enamel shall show no signs of incompatibility such as blistering, wrinkling, or peeling when topcoated with coatings manufactured in accordance with MIL-PRF-24635, type III, MIL-DTL-24607, and MIL-PRF-24596. Visually examine for incompatibility immediately after topcoat application. After 48 hours, visually examine for compatibility and perform and meet 3.6.8 requirements.

3.6.9.2 Compatibility with primers and undercoats. When tested as specified in 4.6.11.2, the enamel shall show no signs of incompatibility such as blistering, wrinkling, or peeling when applied over fully cured or dried films of coatings manufactured in accordance with TT-P-645 or MIL-DTL-24441, type III and type IV formula 150 primer. Visually examine for incompatibility immediately after topcoat application. After 48 hours, visually examine for compatibility and conformance to 3.6.7 requirements.

3.6.10 Surface appearance. When tested as specified in 4.6.12, a flow-out film on glass of the mixed enamel, after drying for 24 hours, shall exhibit a surface smooth in appearance and free of defects, such as loss of dry, color drift, seeding, pinholes, coarse particles, skins, or agglomerates of any kind.

3.6.11 Film hardness. When tested in accordance with 4.6.13, the test enamel shall have a minimum pencil film hardness of HB.

3.7 Directions for use. The directions for the mixing and application of the enamel shall be provided. The directions shall comply with 29 CFR OSHA Hazard Communication Act.

3.8 Batch specific VOC certification. Label instructions shall be prepared in accordance with 29 CFR 1910. Each container shall be affixed with a hazardous chemical warning label in accordance with 29 CFR 1910.1200, subpart Z. To comply with 40 CFR 63, subpart II, the following two statements shall appear on each paint can label:

a. A certification that the paint in the container meets the National Emission Standards for Hazardous Air Pollutants requirements for shipbuilding and ship repair.

b. A statement of the ratio of volatile content to solids expressed as grams of volatile HAPs per liter of solids.

3.9 Ozone-depleting chemicals. The materials used in the enamel shall not contain class I or class II ozone-depleting chemicals as defined in 40 CFR 82.

3.10 Off-gassing. The enamel shall be evaluated for off-gassing in accordance with the requirements of 4.8. Based on the circumstances of use and the chemical nature of the enamel, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the enamel shall be assigned to either the “Permitted” or “Limited” category (see 4.8 and 6.9).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First article inspection. First article inspection shall be performed on a random unit sample drawn from the first production batch of enamel. First article testing shall include all tests specified in 4.6, including those noted as "conformance inspection test".

4.3 Conformance inspection. Conformance inspection shall consist of all tests specified in table VI. At a minimum, conformance inspections shall be performed on the first lot and every 19,000 liters (5,000 gallons) thereafter. A lot shall consist of all components of a coating formulation from a single uniform batch or single uniform blend of batches offered for delivery at one time. Conformance inspection shall be conducted in accordance with method 1031 of FED-STD-141 and shall consist of flash point (ASTM D3278), dry time (ASTM D5895), and condition in container (ASTM D1849).

4.4 Inspection conditions. Unless otherwise specified in the applicable test paragraphs of this specification, all inspections shall be performed in accordance with the test conditions specified in 4.5 and 4.6.

4.5 Test panel preparation. Steel panels [minimum size 60 mm by 90 mm by 3 mm (2.4 inches by 3.6 inches by 0.12 inch) thick [nominal]) shall be prepared for testing by abrasive blasting to a near white metal surface in accordance with SSPC-SP 10/NACE No. 2 to achieve a minimum 0.051 mm (0.002 inch; 2.0 mil) surface profile. The blasted panels shall be solvent wiped to remove any traces of oil or residue, blown dry with dry air, wrapped in plastic and stored in a desiccator, oven, air-tight storage bag, or use other means to prevent rusting prior to coating. The abrasive used for blasting shall be in accordance with MIL-PRF-22262 and be listed on the current qualified products list. Unless otherwise specified in the applicable test paragraphs of this specification, the test panels shall be coated with TT-P-645 primer applied in one coat at a wet film thickness that will yield a dry film thickness (DFT) 50 percent greater than the blast profile. Unless otherwise specified in the applicable test paragraphs of this specification, dry time shall be 48 hours at 21 °C (70 °F).

4.6 Test procedures/methods. Test procedures/methods shall be in accordance with table VI.

TABLE VI. Test procedures.

Test	Requirement	Test	ASTM test method
Toxicity	3.3	4.7	----
Pigment content	Table III	----	ASTM D2698
Volatiles vehicle content	Table III	----	ASTM D2369
Nonvolatile vehicle content	Table III	----	----
Water content 1/	Table III	----	ASTM D1364
VOC g/L 1/	Table III	4.6.1	----
Metal and asbestos content 1/	3.5.1	4.6.2	----
HAP solvents 1/	3.5.2	4.6.3	ASTM E1252
Phthalic anhydride 1/	Table II	4.6.4	----
Rosin/rosin derivative 1/	Table II	----	----
Phenolic resin 1/	Table II	----	----
Soya oil 1/	Table II	----	----
Drying oil acids	Table II	----	----
Coarse particles/skins	Table III	----	ASTM D185
Viscosity 2/	Table III	----	ASTM D562
Weight per liter (gallon) 2/	Table III	----	ASTM D1475
Drying time Dry to touch 2/ Dry through	Table III	----	ASTM D5895

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TABLE VI. Test procedures – Continued.

Test	Requirement	Test	ASTM test method
Fineness of grind [2/]	Table III	----	ASTM D1210
Flashpoint	Table III	----	ASTM D3278
Gloss [2/]	Table III	----	ASTM D523
Sag resistance [2/]	Table III	----	ASTM D4400
Odor	3.6.1	----	ASTM D1296
Color and Surface appearance [2/]	3.6.2 3.6.10	4.6.6 4.6.12	ASTM E1347/D2244 ----
Flexibility	3.6.3	4.6.7	ASTM D522/D522M
Condition in container	3.6.5	4.6.8	ASTM D1849
Storage stability Full container Partially full container	 3.6.5.1.1 3.6.5.1.2	 4.6.9.1 4.6.9.2	ASTM D869/D1849
Rolling properties	3.6.6.1	4.6.15.1	ASTM D5150
Brushing properties	3.6.6.2	4.6.15.1	----
Spraying properties	3.6.6.3	4.6.15.2	----
Cross-cut adhesion test	3.6.7	----	ASTM D3359, method B
Water resistance	3.6.8	4.6.10	----
Compatibility with topcoats with primer and undercoats	 3.6.9.1 3.6.9.2	 4.6.11.1 4.6.11.2	----
Contrast ratio [2/]	Table III	----	ASTM D2805
Pencil hardness	3.6.11 and table III	4.6.13	ASTM D3363
Off-gassing	3.10	4.8	----

NOTES:

[1/] By data certification - Can be referred by manufacturer to the certificate of analysis of received ingredients, as mentioned previously, if product is not described as containing phenolic or rosin modification, then that can be read as a negative. Coating manufacturer may use certificates of analysis for raw materials to satisfy requirements.

[2/] Conformance inspection test (see 4.3).

4.6.1 VOC. VOC content shall be determined in accordance with 40 CFR 60, appendix A-7, method 24. VOC content shall be as specified in table III.

4.6.2 Soluble and total metal content. Asbestos, soluble, and total metal content, with the exception of tantalum and tungsten, shall be determined on a dry film of the enamel in accordance with the 40 CFR 261.24(a) and the appropriate test listed below (see table VII and table VIII). Asbestos shall be analyzed in accordance with 29 CFR 1915.1001, appendix K and the results shall be recorded as percent by weight of the dry enamel film. Soluble metal content shall be analyzed as milligrams per liter (mg/L) (see table IV). Total metal content shall be analyzed as percent by weight of the dry enamel film. The test results for asbestos, each individual soluble metal, and total metal shall be as specified in 3.5.1. Tantalum and tungsten soluble metal content and total metal content shall be analyzed as specified in 4.6.2.1.

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TABLE VII. Test methods for evaluating solid waste physical/chemical methods, EPA SW-846.

Metal/material	Digestion test method
All metals, except chromium (VI)	3050
Chromium (VI)	3060
Metal/material	**SW-846 Analysis test method**
Antimony	7040 or 7041
Arsenic	7060 or 7061
Barium	7080 or 7081
Cadmium	7131
Total chromium	7190
Chromium (VI)	7195, 7196, or 7197
Lead	7421
Mercury	7470 or 7471
Nickel	7520 or 7521
Selenium	7740 or 7741
Silver	7760 or 7761

TABLE VIII. Test methods for chemical analysis of water and waste, EPA 600/4-79-020.

Metal/material	Test method
Beryllium	210.1 or 210.2
Cobalt	219.1 or 219.2
Copper	220.1 or 220.2
Fluoride	340.1, 340.2, or 340.3
Molybdenum	246.1 or 246.2
Thallium	279.1 or 279.2
Vanadium	286.1 or 286.2
Zinc	289.1 or 289.2

4.6.2.1 Tantalum and tungsten content. Determine the tantalum and tungsten content of the enamel using any appropriate spectroscopy test method. Conduct the tests in accordance with the equipment manufacturer's directions for the use of the instrument. Test results for tantalum and tungsten shall be as specified in table IV and table V.

4.6.3 HAP solvents content. HAPs content shall be determined in accordance with EPA Method 311. Solvent fractions shall be identified in accordance with ASTM E1252. Test results shall be reported as percent by weight of the total enamel. Results shall be in accordance with the requirements of 3.5.2.

4.6.4 Phthalic anhydride. Phthalic anhydride content of the nonvolatile vehicle shall be determined (see 6.2) and shall conform to table II.

4.6.5 Drying oil acids. The unsaponifiable, drying oil acids shall be determined (see 6.2) on the isolated vehicle and shall conform to table II.

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4.6.6 Color testing

4.6.6.1 Instrumental color deviation determination. Test specimens shall be prepared in accordance with ASTM D823 Test methods C or E using a nominal 0.006-inch blade film applicator on to a clear plate glass of not less than ⅜-inch (nominal) thickness, which has been ground to a uniform finish with 1F carborundum. A standard black and white Leneta chart is also acceptable. The color deviation shall be determined in accordance with ASTM D2244 using an instrument having a D_{65} light source (10-degree observer) or C (2-degree observer), a 45-degree illumination angle, and a 0-degree viewing angle. The instrument shall be calibrated in accordance with manufacturer instructions and shall be demonstrated to read the color of National Institute for Science and Technology (NIST [formerly National Bureau of Standards; NBS]) traceable standards with color deviations not greater than allowed in 3.6.2. A recommended application of ASTM D2244 follows. After calibration of the instrument, measure the CIE color values of not less than two color number 26307 of SAE-AMS-STD-595 color cards that were received from the Government not greater than 1 year prior to the date of this use. Make at least three readings per card taken at 1-inch intervals on the long axis, 1 inch from the edge. Calculate ΔE for each color card. Determine the mean values of ΔE and the CIE color values for the color cards. If any individual CIE color value differs from the mean color value by more than 0.05, use a different color card and recalculate the mean values. These mean values will be used to calculate the color difference values. Measure the CIE color values of the test enamel using the values of three readings taken at 1-inch intervals on the long axis, 1 inch from the edge. Calculate the mean CIE color values of the test enamel. Calculate the color difference terms. If using an instrument that automatically calculates the color differences, enter the color card mean values as the base line for the color difference calculations. The results of the color difference calculations shall be in accordance with the requirements of 3.6.2.

4.6.6.2 Referee color matching. If the enamel measured in accordance with 4.6.6.1 does not match the color card when visually examined, an independent laboratory certified to the NIST National Volunteer Laboratory Accreditation Program for paint testing shall resolve the discrepancy. When prepared and tested as in 4.6.6.1, the color shall be in accordance with the requirements of 3.6.2.

4.6.6.3 Alternative color measuring equipment. Alternative color measuring equipment may be proposed to the Technical Authority or GSA by demonstrating that the particular instrumentation to be used measures the color parameters with the accuracy needed for color deviation. The alternate instrumentation shall be able to provide color parameters that result in the color defined in 3.6.2 and the correlation factors for converting the instrument's color parameters into those of equipment meeting 4.6.6.1 shall be established by measuring an acceptable Navy Formula no. 111 enamel previously measured on an instrument meeting the requirements of 4.6.6.1.

4.6.7 Flexibility. Flexibility shall be determined in accordance with ASTM D522/D522M. The enamel shall be applied to a flat metal plate of approximately 31 gauge coated with a DFT of 0.025 mm (0.001 inch +0.0003 inch) of TT-P-645 primer by means of a doctor blade capable of yielding a DFT of 0.025 mm (0.001 inch+0.0003 inch). The panel shall be allowed to air-dry for 2 hours then baked for 24 hours at 100 to 105 °C (212 to 221 F). The panel shall then be removed from the oven and allowed to stand for 30 minutes at laboratory temperature [nominal 23 °C (73 °F)]. The panel shall then be bent over a 3.2-mm (⅛-inch) mandrel or a tapered mandrel where the minimum diameter is 3.2-mm (⅛-inch) and the film examined at the bent under a magnification of five diameters. Flexibility shall be as specified in 3.6.3.

4.6.8 Condition in container. A container of the product in the size ordered shall be inspected. The product shall be equilibrated to test laboratory conditions [nominal 23 °C (73 °F)]. Evidence of pressure or vacuum in the unopened container shall be noted. The container shall then be opened and examined in accordance with ASTM D1849 for evidence of skinning, corrosion of container interior, odor of putrefaction, rancidity or souring, gel bodies, and hard settling. If the sample is in a 1-liter (1-quart) or smaller container, the character of the lower (or settled) layer shall be determined with a spatula as specified in ASTM D869. If the sample is larger than 1 liter (1 quart), this step shall be omitted. The enamel shall be hand-stirred 300 stirs in 2 minutes with a spatula appropriate to the container, stirring so as to ensure uniform distribution of any settled material. Immediately after stirring, the consistency of the enamel shall be measured using the Krebs-Stormer viscometer. The enamel shall be applied to a test panel. and after it has dried, the enamel film shall be examined for grains 0.8 mm (1/32-inch) in diameter, even larger gelatinous lumps, and streaks caused by such grains or lumps. The condition of the product shall be as specified in 3.6.5.

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4.6.9 Storage stability.

4.6.9.1 Full container. Storage stability shall be tested in accordance with ASTM D1849. A container of the product of the size ordered shall be stored at the temperature recommended in the manufacturer's data sheet or ASTM F718 for a period of 1 year. After completion of 1 year, the stored sample shall be equilibrated to test laboratory conditions [nominal 23 °C (73 °F)]. Evidence of pressure or vacuum in the unopened container shall be noted. The container shall then be opened and examined in accordance with ASTM D1849 for evidence of skinning, corrosion of container interior, odor of putrefaction, rancidity, or souring. If the sample is in a 1-liter (1-quart) or smaller container, the character of the lower (or settled) layer shall be determined with a spatula as specified in ASTM D869. If the sample is larger than 1 liter (1 quart), this step shall be omitted. The enamel shall be hand-stirred 300 stirs in 2 minutes with a spatula appropriate to the container, stirring so as to ensure uniform distribution of any settled material. Immediately after stirring, the consistency of the enamel shall be measured using the Krebs-Stormer viscometer. The enamel shall be applied to a test panel and after it has dried the enamel film shall be examined for grains 0.8 mm (1/32 inch) in diameter, even larger gelatinous lumps, and streaks caused by such grains or lumps. The condition of the product shall be as specified in 3.6.5.1.1.

4.6.9.2 Partially full container. Skinning shall be determined after 48 hours in accordance with ASTM D1849, except that a ¾-filled ½-pint, multiple friction top can shall be used. The can shall be resealed and aged for 7 days at 60 °C (140 °F). Results shall be in accordance with the requirements of 3.6.5.1.2.

4.6.10 Water resistance. The enamel shall be sprayed on a vertically oriented, solvent-cleaned, cold-rolled steel panel prepared as specified in 4.5 that is a minimum of 100 mm (4 inches) by 300 mm (12 inches), to a DFT of 0.051- to 0.076-mm (2- to 3-mils). Allow panel to dry for at least 48 hours at laboratory conditions [21±2 °C (70±5 °F) and 50±5-percent relative humidity]. Immerse the lower six inches of the panel in room temperature distilled water for 24 hours. The condition shall be as specified in 3.6.8 and 3.6.10.

4.6.11 Compatibility. Compatibility of the enamel with topcoats and primers shall be tested in accordance with 4.6.11.1 and 4.6.11.2, respectively.

4.6.11.1 Compatibility with topcoats. Prepare and prime three panels as specified in 4.5. To the first panel apply a 0.051- to 0.076-mm (2- to 3-mil) DFT topcoat of MIL-PRF-24635, type III; to the second panel apply a 0.051- to 0.076-mm (2- to 3-mil) DFT topcoat of MIL-DTL-24607; to the third panel apply a 0.051- to 0.076-mm (2- to 3-mil) topcoat of MIL-PRF-24596. Allow test panels to dry at least 48 hours at laboratory conditions [21±2 °C (70±5 °F) and 50±5 percent relative humidity]. To each of the topcoated panels, apply one coat of the test enamel at 0.051- to 0.076-mm (2- to 3-mils) DFT. Allow test panels to dry at least 48 hours at laboratory conditions [21±2 °C (70±5 °F) and 50±5 percent relative humidity]. Evaluate cross-cut adhesion 48 hours after topcoat application by test method B of ASTM D3359. Enamel condition shall be as specified in 3.6.7 and 3.6.9.1.

4.6.11.2 Compatibility with primers. Prepare and prime three panels as specified in 4.5. To the first panel apply a 0.051- to 0.076-mm (2- to 3-mils) primer coating TT-P-645; the second panel with 0.051- to 0.076-mm (2- to 3-mils) of MIL-DTL-24441, F.150, type III paint and to the third panel apply 0.051- to 0.076-mm (2- to 3-mils) of MIL-DTL-24441, F.150, type IV paint. Apply one coat at 0.051- to 0.076-mm (2- to 3-mils) DFT of the test enamel. Examine for blistering, wrinkling, or peeling of the enamel immediately after topcoat application and 48 hours after topcoat application. Check adhesion 48 hours after topcoat application by test method B of ASTM D3359. Enamel condition shall be as specified in 3.6.7 and 3.6.9.2.

4.6.12 Surface appearance. Prepare a flow-out film of the enamel by pouring approximately 15 mL of the mixed enamel across a glass panel near the upper edge, while the panel is lying flat. Then tilt the panel to allow the coating to spread over all but the upper edge. Next, place the panel in an almost vertical position and allow to drain. After 24 hours, examine the film for compliance with 3.6.10. Coarse particles, skins, and agglomerates are characterized by being larger than the dispersed pigment in particle size and extending beyond the plane of the film.

4.6.13 Film hardness. To a steel panel prepared and primed in accordance with 4.5, apply 0.051- to 0.076-mm (2- to 3-mils) of the test enamel and dry for seven days at laboratory conditions [21±2 °C (70±5 °F) and 50±5-percent relative humidity]. Test in accordance with table VI and evaluate for compliance with 3.6.11.

4.6.14 Fineness of grind. The fineness of grind test shall be conducted in accordance with the test method prescribed by table VI. Results shall be in accordance with the requirements of table III.

4.6.15 Application properties. Application properties of the coating under usual application techniques shall be tested in accordance with 4.6.15.1, 4.6.15.2, and 4.6.15.3.

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4.6.15.1 Rolling properties. Enamel shall be tested in accordance with ASTM D5150, using a 75-mm (3-inch) roller with a 4.8- to 6.4-mm nap (3⁄16- to ¼-inch) that has been pre-wetted with the coating to be applied, apply 0.051 to 0.076 mm (2 to 3 mils) of the test enamel to a vertically oriented, 6 sqft practical opacity chart that aligns with ASTM D5150. Only a quantity of enamel sufficient to cover 3 sqft at 2 to 3 mils should be utilized. Allow to dry at least 24 hours at laboratory conditions [21±2 °C (70±5 °F) and 50±5-percent relative humidity]. Evaluate for compliance with 3.6.6.1.

4.6.15.2 Brushing properties. The enamel shall be brush applied to a vertically oriented, solvent-cleaned, cold-rolled steel panel that is a minimum of 600 mm (24 inches) by 600 mm (24 inches), using a brush (50- to 75-mm [2- to 3-inches] wide) that has been pre-wetted with the enamel to a DFT of 0.051- to 0.076-mm (2- to 3-mils). The enamel shall be capable of being brushed out and laid off without excess drag on the brush. Allow test panel to dry at least 24 hours at laboratory conditions [21±2 °C (70±5 °F) and 50±5-percent relative humidity]. Evaluate for panel for sags, runs, and brush marks in accordance with method 4321.3 of FED-STD-141 for compliance with 3.6.6.2.

4.6.15.3 Spraying properties. The enamel shall be sprayed on a vertically oriented, solvent-cleaned, cold-rolled steel panel that is a minimum of 100 mm (4 inches) by 300 mm (12 inches), to a DFT of 0.025- to 0.051-mm (1- to 2-mils). Allow panel to dry for at least 24 hours at laboratory conditions [21±2 °C (70±5 °F) and 50±5 percent relative humidity]. Evaluate wet and dry condition for compliance with spraying properties in accordance with method 4331.2 of FED-STD-141 to determine compliance with 3.6.6.3.

4.7 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.3 as specified (see 6.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the enamel using data provided by the manufacturer/distributor to the NMCFHPC (see 3.3 and 6.8).

4.8 Off-gassing. The enamel shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled “Material Control Program” (see 3.10 and 6.9). If the Navy determines that off-gas testing is required, testing shall be conducted at a test facility approved by the Technical Authority (see 3.10). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.10).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The enamel covered by this specification is intended for use as a finish or topcoat on equipment, machinery, furniture, and electrical equipment in marine service where fire-retardancy is not required. This product may be applied where air pollution regulations apply. This enamel is a non-lead, non-chromate, non-asbestos material. The dried paint film has been tested to U.S. EPA standards and the paint debris is, as of the time of this specification, a non-hazardous waste. It may be applied by brush or spray and is suitable for any metal or wood structure. During development, this enamel was easily sprayed with a Binks #7 HTES (high transfer efficiency special) gun. Spraying with a high volume, low pressure (HVLP) gun was successful using an AIRVERTER® HVLP Sprayer available from Smith Eastern Corporation. The basic formulation provides an enamel of a gray color. This enamel has been formulated to comply with air pollution regulations which allow a maximum VOC of 250 grams of solvent minus water per liter of paint (2.08 lb/gal) as delivered, and does not require thinning for brush, roller, or spray application. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained and approved by the Technical Authority.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article is required (see 3.1).

c. Requirements for an HHA (see 3.3, 4.7, and 6.8.1).

d. When a certificate of conformance is required to determine %WT (see 4.6.4 and 4.6.5).

e. Packaging requirements (see 5.1).

f. Whether Safety Data Sheets (SDS) are required with each shipment (see 6.6).

g. Recommendation for manufacturer's commercial product data sheet for use in accordance with ASTM F718 (see 6.7).

6.3 Technical Authority approval and direction. Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures will be submitted to the Technical Authority for evaluation and direction. Requests should include supporting documentation.

6.4 First article inspection. First article inspection is required for all manufacturing sources. The contracting officer should specify to offerors that the item is a first article sample from the first production batch (see 3.1) to be tested. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspections to those bidders offering a product that has been previously tested by the Government, and that bidders offering such product, who wish to rely on such tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Volatile content. Although the container marking specifically refers to the Federal regulations, the paints may be used anywhere else a product complying with 3.2 is allowed. This includes other air pollution control districts or similar areas controlling the emission of solvents into the atmosphere.

6.6 SDS. The contracting activity should be provided a SDS at the time of contract award. The SDS should be provided in accordance with OSHA section 1910.1200, 29 CFR chapter XVII and found as part of FED-STD-313. OSHA section 1910.1200 requires reporting threshold criteria for known or suspected human carcinogens on SDS 0.1 percent or greater, and 1 percent or greater for other health hazards. The SDS should be included with each unit of issue of material covered by the specification, when specified (see 6.2). Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Directions for use. The manufacturer should provide written directions on each container for the mixing and application of the enamel supplied and this direction should include all information necessary to comply with OSHA Hazard Communication Act and FED-STD-313. In addition, the manufacturer should prepare an ASTM F718 data sheet (see 6.2), which should separately detail requirements for small unit (pint, quart, gallon) and large unit (five-gallon) containers.

6.8 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

6.8.1 Toxicological product formulations. When specified (see 6.2), approval by the Technical Authority is required to conduct an HHA (see 4.7). The Technical Authority will request the HHA, evaluate the HHA, and provide the HHA back to the manufacturer. The manufacturer will retain the applicable files or data package showing compliance with First Article requirements to be provided upon request to paint procuring activities. To minimize requirements for separate approval action by the Technical Authority, contract ordering data should include information related to this request, whenever possible.

6.9 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects

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on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, NAVSEA S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to the Naval Sea Systems Command, SEA 05S, 1333 Isaac Hull Ave., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas test is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Ave., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.10 Suggested packaging. Suggested packaging recommendations are contained in table IX and table X.

TABLE IX. Suggested packing and packaging.

Packaging	Recommendations for direct Government acquisitions
Unit of procurement	The coatings covered by this specification should be purchased by volume. The unit of procurement should be in multiples of 1 liter or 1 U.S. liquid gallon at 15.5 °C (60 °F). Volumes less than 1 liter or 1 U.S. liquid gallon furnished in cartridges, packs, and aerosol cans should be purchased as defined in the acquisition document.
Containers	a. The paint should be furnished in cans capable of holding 0.47 liter (1 pint), 0.945 liter (1 quart), 3.78 liters (1 gallon), or 18.9 liters (5 gallons). In addition, paint may also be furnished in cartridges, burst packs, and aerosol cans as defined in the acquisition document. b. Friction plug containers should be in accordance with PPP-C-96, type V, class 2. Interior coatings should be as specified therein. Exterior coatings, including side seam stripping, should be as specified therein for plan B. Wire handles as specified therein, should be provided for the 1-gallon container. Closure of the properly filled and sealed cans should be as specified in the appendix thereto. c. Pails should be to PPP-P-704. d. All containers should comply with the requirements of the Uniform Freight Classifications (UFC), the National Motor Freight Classification (NMFC), and the applicable requirements of the Code of Federal Regulations 49 CFR, Department of Transportation (DOT).
Intermediate containers	a. Paints should be packaged in intermediate containers. b. Intermediate containers should be close-fitting corrugated fiberboard boxes in accordance with UFC, NMFC, and 49 CFR requirements. Fiberboard used in the construction of interior (unit and intermediate) and exterior containers, including interior packaging forms, should conform to the best commercial practice. Such classes should be domestic fire-retardant or weather resistant fire-retardant as specified.
Commercial packaging	a. Commercial packaging should be to ASTM D3951. b. All containers should comply with the requirements of the UFC, the NMFC, and the applicable requirements of 49 CFR, DOT.

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TABLE IX. Suggested packing and packaging – Continued.

Packaging	Recommendations for direct Government acquisitions
Packing	Packing should be specified as follows: a. Overseas delivery (level A) packing. Intermediate containers of paint should be packed in close-fitting wood boxes conforming to the best commercial practice. Box closure and strapping should be as specified in the applicable box specification or the appendix thereto except that strapping should be flat and the finish B. b. Domestic delivery (level B) packing. Level B packing should be as for level A, except that boxes should be domestic type or class and the strapping should be finish A or B. c. Commercial packing. The paint, in the specified unit and intermediate containers should, as applicable, be packed in multiples of like sizes in accordance with UFC, NMFC, and 49 CFR requirements.
Palletization	Intermediate containers should be palletized in accordance with the best commercial practice. Only one size unit or intermediate container should be placed on a pallet.
Packing for Navy acquisitions	Treated lumber and plywood. All lumber and plywood, including laminated veneer materials, used in shipping container and pallet construction, member, blocking, bracing, and reinforcing should be fire-retardant treated material in accordance with MIL-L-19140 as follows: a. General use, weather resistant: MIL-L-19140, type II, category I. b. General use, non-weather resistant: MIL-L-19140, type I, category I.
SDS and ASTM F718	A copy of the SDS and ASTM F718 should be attached to the shipping document for each destination (see 6.6).
VOC certification sheets	VOC certification sheets for each batch of coating will be provided when requested by the procuring activity.

TABLE X. Suggested marking.

Marking type	Recommended marking
Bar codes	Marking should include bar codes.
Hazardous warnings	a. Labels should be in accordance with 29 CFR parts 1910, 1915, 1917, 1918, 1926, and 1928. b. All individual containers should have the following marking: "CAUTION: This paint contains volatile solvents, with probable hazardous vapors. Use with adequate ventilation. Avoid prolonged breathing of vapors or spray mists. The solvents are highly flammable, avoid open flame, and smoking." c. Each paint container, shipping container, and palletized load should be marked with the appropriate hazardous symbol in accordance with FED-STD-313.
VOC content	VOC should be stated on the label using the following statement: "Contains (insert VOC content) grams per liter (insert VOC content in pound/gallon) of volatile organic compound per 40 CFR 60, chapter 1, appendix A-7, method 24."
OSHA Hazard Communication Act and FED-STD-313	Markings should include all information necessary to comply with OSHA Hazard Communication Act and FED-STD-313.
Mixing and use instructions	Directions should include mixing, application equipment directions, limitations on thinning, temperature range for use and surface preparation recommendations. Directions should refer user to data sheets, SDS, and ASTM F718 for information.

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6.11 Definitions.

6.11.1 Lot. A lot consists of all material manufactured at one time (batch) or all material produced by mixing multiple batches into one uniform unit of material.

6.12 Subject term (key word) listing.

Alkyd
Asbestos
Cobalt compounds
Equipment paint
Paint
Pigment
Shipboard surfaces
Topcoat
Volatile organic compound

6.13 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians
Army – CR4
Navy – SH
Air Force – 99

Review Activity:
Misc – DS

Preparing activity:
Navy – SH
(Project 8010-2025-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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